Insurance Document Intelligence for Automated Policy and Claim Intake

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Abstract— This paper describes a local document-intake prototype combining OpenCV preprocessing, Tesseract OCR, rule-based document classification and candidate field extraction, cross-document consistency checks, and a Streamlit review dashboard. The project compares OCR over independent image variants and three Tesseract page segmentation modes (PSM 3, 6, and 11). On a small copied synthetic insurance sample, 23 scans were processed and 19 had same-stem PDF text references; four police-report scans had no paired reference. An independent audit reproduced all 627 scored CER/WER pairs from saved predictions. Because the PDF text was not exhaustively aligned to scan pixels, the OCR scores are provisional. CLAHE had lower aggregate error than Original in the reported sample comparisons, but no significance test or independent held-out evaluation was performed. The system surfaces candidate evidence for human review and does not make claim decisions.

Index Terms— document image processing, OCR, CLAHE, Tesseract, CER, WER, insurance intake prototype.

I. INTRODUCTION

Scanned intake pages can include uneven illumination, skew, low contrast, compression, and diverse layouts. Text recognition can support document review, but raw OCR output is not an authoritative record. A useful academic prototype should expose its image transformations and OCR output, connect extracted candidates to supporting text, and preserve human review. This work joins those steps in a local modular application. It studies established image operations on a copied synthetic sample and documents the limits of the available reference text. Its claims are deliberately narrow: the work is an integration and descriptive comparison, not a new OCR algorithm or a production insurance system.

II. SYSTEM ARCHITECTURE

The Streamlit application calls shared Python modules. The ingestion layer loads images or rasterizes PDFs. *process_document()* invokes preprocessing and OCR, records per-page outputs, and allows downstream extraction to use a confidence-ranked variant. The experiment independently scores each saved image-variant/PSM prediction, avoiding substitution of the application-selected view into all measurements. Classification uses an explainable rule baseline. Field extraction uses regular expressions and attaches OCR evidence and missing statuses. Cross-document rules compare repeated values and flag mismatches or manual review needs. JSON artifacts hold detailed records and SQLite indexes local history. An optional language-model summary is not required for core functionality. No component approves, denies, or determines insurance coverage.

III. RELATED TECHNICAL CONTEXT

Otsu's method selects a gray-level threshold by histogram class separation [1]. CLAHE performs local contrast enhancement with a clip limit to control amplification [2]. Morphology applies shape operations using a structuring element; parameter scale can damage or merge character strokes. Tesseract supports page segmentation assumptions: automatic layout, a uniform text block, and sparse text search (PSM 3, 6, and 11) [3]. FUNSD provides a separate form-understanding benchmark [4]; it is not an insurance dataset and is not mixed into the insurance experiment.

IV. DATA AND EXPERIMENTAL METHOD

The local inventory contains 88 records from the Strata sample manifest and 10 Gritworks PDFs. The Strata sample is synthetic and not representative of real-world insurance data. From it, 23 eligible document scans were processed; 19 matched to a same-stem PDF and four police-report images did not. Gritworks was inventory-only. The OCR experiment used Tesseract 5.5.3 and PSM 3, 6, and 11 on the same images and reference text. For every page it evaluated original, grayscale, CLAHE, histogram equalization, global threshold, Otsu threshold, adaptive threshold, Gaussian denoising, median denoising, morphology, and deskewed grayscale. Deskew records the estimated angle and rotation settings; it is evaluated independently, without assuming benefit.

Same-stem PDFs served as provisional text references. PyMuPDF extracted page text with sorting enabled; scoring normalization applied Unicode NFKC, casefolding, and whitespace collapse while retaining punctuation. The extracted layer was not exhaustively visually adjudicated against the scan. Changing reading-order options affected 10 of 19 paired normalized references in the audit. Four unpaired images remain OCR processed but are excluded from reference metrics. No training split was used for insurance testing, and there is no independent held-out corpus. The separate FUNSD exercise used its 50-page test split only.

CER is character edit distance divided by the number of reference characters. WER is word-token edit distance divided by reference word count. Macro scores average normalized document scores, weighting each document equally; micro scores pool edit counts and reference denominators, weighting longer references more. Empty references were explicitly considered; the saved scored set contained no empty references. Field precision/recall/F1 use strict normalized exact matching against synthetic metadata and apply only where reference support is defined.

V. RESULTS

Table I summarizes Original and CLAHE macro scores. Lower is better within this provisional sample comparison. An independent audit recomputed all 627 of 627 scored CER/WER values from saved predictions and references. Each PSM/variant comparison scored 19 paired documents; four unmatched scans were excluded. No duplicate evaluation keys were found.

PSM	View	Macro CER	Macro WER
3	Original	0.306785	0.377461
3	CLAHE	0.177384	0.271460
6	Original	0.329215	0.421155
6	CLAHE	0.224573	0.328093
11	Original	0.287972	0.377817
11	CLAHE	0.142398	0.248395

TABLE I. PROVISIONAL MACRO OCR RESULTS; 19 PAIRED DOCUMENTS PER CELL.

CLAHE had lower CER on 18/19, 17/19, and 19/19 documents for PSM 3, 6, and 11, respectively; these sample counts are descriptive. The metric audit verifies arithmetic, not that PDF text faithfully represents pixels in each scan.

VI. FIELD EXTRACTION AND ERROR CONTEXT

For PSM 3 Original, 100 of 124 applicable metadata values matched exactly (precision 0.9091, recall 0.8065, F1 0.8547 under the reported strict micro scheme). Twenty-four nonmatches were classified as 3 incorrect, 12 missing, 9 reference ambiguity, and 0 unavailable. The reference-supported subset has a different denominator and is reported separately in the audit. These metadata comparisons are distinct from OCR CER/WER and depend on synthetic entity joins and field applicability.

Macro and micro scores answer different questions. Macro aggregation assigns the same weight to each document and reveals typical per-document normalized error. Micro aggregation pools edit distances and denominators, so long references contribute more. Large disagreement can arise when shorter documents have higher normalized error or when reference lengths vary. In the saved report, all comparisons have a common pooled reference denominator for a given metric/configuration. Neither aggregation resolves reference-text or reading-order error.

OCR errors can be interpreted as substitutions, insertions, and deletions through edit-distance alignment. Word-level alignment can show missing/extra tokens and segmentation differences, although token boundaries make some OCR spacing changes appear as multiple edits. CER/WER do not by themselves distinguish visual recognition failure from a ground-truth ordering discrepancy. The project audit reports reading-order sensitivity and retains the original reference construction rather than silently rewriting text.

VII. IMAGE-PROCESSING VISUAL EVIDENCE

The accompanying submission folder contains actual preprocessing and OCR comparison sheets produced from the existing synthetic sample and shared processing pipeline. They show original, grayscale, CLAHE, thresholded, denoised, morphological, and deskewed views alongside saved OCR excerpts. Figures are inspection aids, not separate accuracy claims. The deskew variant estimates a page angle and records interpolation and border handling. Its inclusion does not imply that correction improves OCR; individual deskew regressions occur in the saved evaluation.

VIII. DASHBOARD AND IMPLEMENTATION

The local dashboard has eight sections: Overview, Document Upload, Image Processing Lab, OCR Results, Document Intelligence, Claim Validation, AI Summary, and Evaluation. Upload-driven sections depend on a processed document. Candidate values and consistency outcomes are review cues; confidence is not calibrated. Results can be stored as JSON artifacts and indexed in SQLite. Tesseract is configured centrally, and the shared pipeline is reused by application and experimental scripts. The test suite contains 27 tests, all passing under the workspace-permitted temporary directory.

The evaluation page reads the latest saved experiment record and exposes aggregate OCR comparisons across PSM and preprocessing variants; it also reports paired reference counts, unmatched images, and OCR failures. The interactive review pages depend on app state: without uploaded documents they show empty-state prompts. For actual use, the document uploader accepts raster images and PDFs; processing is local, and the configured engine and extracted fields are shown in subsequent pages. These interface capabilities describe implemented behavior only, not operational insurer use.

IX. FIELD METRIC INTERPRETATION

An exact-match field score rewards a predicted value only when it matches the normalized metadata value. A wrong nonempty prediction can contribute a false positive and a false negative; a missing prediction contributes a false negative. Precision therefore measures reliability among produced candidate values, while recall measures the share of reference-supported values recovered. F1 is their harmonic mean. An unavailable or ambiguous ground-truth value should not be silently marked wrong. This project's audit categorizes such cases separately and reports a reference-supported subset. Field metrics should not be conflated with text-level CER/WER or used to imply decision quality.

X. LIMITATIONS AND REPRODUCIBILITY

The insurance sample is small and synthetic. Repeated claim and policyholder entities mean scans are not independent. There is no supplied train/development/test split for this sample. PDF selectable text may omit, reorder, or differently space visible content; 4 scans have no paired PDF. As a result, reported OCR comparisons are provisional and cannot establish generalization. The test comparisons are descriptive; no statistical significance test was performed. Configuration choices must not be presented as unbiased final performance after selection on this sample. The prototype is not production-ready, does not represent a real insurer deployment, and does not determine claim outcomes.

To reproduce on Windows PowerShell, create and activate Python 3.12 virtual environment, install *requirements.txt*, install Tesseract OCR 5.5.3 separately, set *$env:TESSERACT_CMD* if needed, then run *python scripts/inspect_insurance_dataset.py*, *python scripts/run_insurance_ocr_experiment.py*, *python scripts/audit_insurance_experiment.py*, *python scripts/generate_academic_visuals.py*, and *python -m pytest -q*. Audit output is timestamped; preserve the report's referenced historical run when reproducing. Do not rerun legacy writers over historic CSV paths.

Threats to validity fall into three categories. Construct validity is limited because a selectable PDF text layer is only a proxy for a human-verified transcription; text order, spacing, OCR normalization, and scan alignment affect edit counts. Internal validity is constrained by repeated entities and the absence of an independently selected held-out sample. External validity is limited by the small synthetic corpus, its layout and image-quality range, and the four unpaired police-report scans. These concerns are not repaired by recomputing edit distance correctly. The audit therefore distinguishes verified score arithmetic from provisional interpretation of those scores.

For future evaluation, an adjudicated image-to-transcript reference should be prepared before comparing configurations. Documents from the same claim or policyholder should remain within one data partition. The preprocessing and PSM configuration should be frozen using development data before final test measurement. Field applicability should be labeled by a reviewer, with ambiguity and unavailability kept distinct from a wrong extraction. Broader data access should be lawful and documented. These are future validation steps, not completed results of this prototype.

XI. CONCLUSION

This project demonstrates a local workflow connecting established image-processing operations, configurable OCR, rule-based candidate extraction, validation cues, saved artifacts, and human review. The audit independently verified the arithmetic of saved CER/WER comparisons and identified reference and denominator limitations. CLAHE had lower reported aggregate error than the original view in this sample, but that observation remains provisional and descriptive. The next scientifically useful steps are scan-level transcript adjudication, clear field applicability labels, entity-level data partitions, a frozen evaluation protocol, and evaluation on appropriately licensed and more representative data. Such steps are necessary before drawing general performance conclusions.

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